

Phase 1: Foundation & Environment

The bare minimum needed to have a physical space to play in.

Task Name	Description (< 30 words)	Dependencies
1.1 Input System Setup	Install Unity's New Input System. Map controls for 4-way movement, shoot, and catch actions.	None
1.2 Base Field Geometry	Create the floor and invisible wall boundaries to prevent the ball from flying out of bounds.	None
1.3 Zone Triggers	Create trigger boxes for the 2 Shooter zones at the ends and the 1 Dodger zone in the middle.	1.2
1.4 Zone Logic	Write a script that restricts players from leaving their designated zones.	1.1, 1.3
1.5 Camera Rig	Position a top-down or isometric orthographic camera to view the entire field perfectly.	1.2

Phase 2: Prototyping Core Mechanics (Placeholders)

Using basic shapes (cubes/spheres) to make the game playable before making it pretty.

Task Name	Description (< 30 words)	Dependencies
2.1 Player Movement Script	Code 4-way movement logic applying forces or velocities to Rigidbody capsules.	1.1, 1.2

2.2 Ball Physics Setup	Create a sphere with a Rigidbody and bouncy physics material.	1.2
2.3 Basic Throw Logic	Script allowing a shooter to instantiate/grab the ball and add forward physical force to it.	1.1, 2.1, 2.2
2.4 Hit Detection System	Script on the ball to detect if it collides with a Dodger capsule.	2.2, 2.3
2.5 Basic Dodge Logic	Script to detect if a ball passes the Dodger without hitting them (raycasts or passing a Z-axis threshold).	2.2, 2.4

Phase 3: The Precision Bar & Advanced Actions

Adding the skill-based mechanics you requested.

Task Name	Description (< 30 words)	Dependencies
3.1 Precision Bar UI	Build a UI Canvas slider with red, yellow, and green colored zones.	None
3.2 Precision UI Animation	Script the slider needle to move rapidly back and forth when an action is triggered.	3.1
3.3 Shooter Accuracy Logic	Link the Precision Bar stop-value to the ball's trajectory offset (green = straight, red = wide).	2.3, 3.2

3.4 Shooter Power Logic	Script a "Hold button to charge" mechanic that increases the ball's velocity upon release.	2.3, 3.3
3.5 Dodger Catch Logic	Trigger Dodger's Precision Bar when ball is near. Green = successful catch, Red = hit.	2.4, 3.2
3.6 Ball Drop/Throw Logic	Allow Dodger to drop the caught ball or throw it back (using the same throw logic).	2.3, 3.5

Phase 4: The Game Loop & Scoring

Turning the mechanics into an actual match.

Task Name	Description (< 30 words)	Dependencies
4.1 Team Manager System	Script arrays/lists managing two teams of 5 players, tracking who is active and who is eliminated.	None
4.2 Spawn Manager	Script to spawn 2 Shooters at the ends and 1 Dodger in the middle at round start.	1.3, 4.1
4.3 Score Manager	Script to add +10 points for a catch and +1 point for a dodge to the active team's score.	2.5, 3.5
4.4 Elimination Logic	When a Dodger is hit, destroy/hide their object and spawn the next player in the array.	2.4, 4.1, 4.2

4.5 Team Swap Logic	When all 5 Dodgers are eliminated, reset the field, swap team roles, and respawn.	4.1, 4.4
4.6 Match End Logic	Detect when both teams have played both roles, compare scores, and declare the winner.	4.3, 4.5

Phase 5: Multiplayer & AI

Allowing people (or computers) to play the game.

Task Name	Description (< 30 words)	Dependencies
5.1 Multiplayer Setup (Local/Net)	Implement split-screen device routing OR a networking framework (like Mirror/Netcode) to sync player positions.	2.1
5.2 PvP State Sync	Ensure the Precision Bar timing, Ball Physics, and Scores sync accurately across the network/controllers.	3.2, 4.3, 5.1
5.3 AI State Machine	Create basic AI logic (Idle, Seek Ball, Aim, Throw, Flee) for testing or single-player mode.	2.1, 2.3
5.4 AI Precision Simulation	Give the AI a random number generator to simulate them stopping the Precision Bar at varying accuracies.	3.3, 5.3

Phase 6: Art, 3D Models & Animation

Replacing the cubes and spheres with actual game art.

Task Name	Description (< 30 words)	Dependencies
6.1 Character Modeling	Create or download a 3D humanoid character model.	None
6.2 Character Rigging	Add a skeleton (armature) to the 3D model so it can move.	6.1
6.3 Environment Modeling	Create high-quality 3D assets for the stadium/field (bleachers, floor texturing, walls).	None
6.4 Animation: Locomotion	Create Idle, Run Forward, Back, Left, and Right animations.	6.2
6.5 Animation: Actions	Create Throw, Catch, and Hit/Fall animations.	6.2
6.6 Animator Controller	Build the Unity Animator tree to blend between running, throwing, and falling based on script variables.	2.1, 6.4, 6.5
6.7 Model Integration	Replace the placeholder capsules with the fully animated 3D models.	2.1, 6.6

Phase 7: Polish, UI, & Audio

Making the game feel complete and satisfying.

Task Name	Description (< 30 words)	Dependencies
7.1 In-Game HUD	Create UI text for current scores, active players remaining, and current team roles.	4.3, 4.4
7.2 Trail VFX	Add a Trail Renderer to the ball so players can easily track its high-speed movement.	2.2
7.3 Impact VFX	Create a particle effect that triggers when the ball hits a player or a wall.	2.4
7.4 Audio Implementation	Import and trigger sounds for footsteps, throwing, catching, ball impacts, and crowd cheers.	6.6, 7.3
7.5 Match End UI	Create the final "Team X Wins!" screen with final scores.	4.6
7.6 Confetti VFX	Build a colorful particle system that explodes on the screen when the Match End UI triggers.	7.5